

LUMENCORE

Portfolio Review Brief | October 4, 2026

Science, geometry and imagination - connected by reproducible evidence.

Founder: Robert Ashworth

LumenCore combines implemented software, quantitative research, evidence tools, funding workflows, virtual engineering and geometric design. The commercial entry point is a bounded buyer-owned evaluation: freeze an authorized question, data and baseline; compare candidates; retain failures; deliver an inspectable decision record.

Working scope: OPEN. The first edition was a selected sample. The expanded review restores substantial omitted work. Disconnected Windows/iCloud roots, private runtimes, unresolved source packages and the wider image archive remain in the recovery queue.

The implemented estate

Recovered work	Evidence and boundary
Mission Control / Quant Lab / Grant Lab	Full historical source recovered. Current redirect or HOLD pages do not erase the interfaces. Current runtime operation needs its own evidence.
Grant Factory / LumenGov	Discovery, qualification, narrative and budget generation, evidence packaging, preflight, immutable approval snapshots and tracking. Drafts and official submissions have distinct stages.
Trading and source pipelines	Research orchestration, strategy sweeps, source registries, paper controls and evidence ledgers. The source review does not establish a profitable edge or current live operation.
LumaJet	A 1,400-scenario generated v2 result shows a tiny positive internal score difference with about 5.08 times baseline planner expansions. Adverse v1 retained; no flight/hardware validation.
EchoForm / LumaSkin	Conventional memory-boundary checks and a 200,000 synthetic-packet endurance receipt. These are implemented research assets; no harmonic/fractal advantage or human/hardware testing inferred.
Immersive and systems work	Unity/Node-RED integration, QMPL, Agent Arena and further engineering sources recovered. Source presence, current execution and external acceptance remain separate.

The June 10 source registry contains **1,481,321 dated row references**, with 12 measured-file-match entries among 17 enabled entries. These are not deduplicated or current live-feed counts. The frozen EIA input was directly recounted at 14,704 rows in this review. Different dates and denominators are not added together.

Archived research source was also recovered: **2,456 reachable commits and 1,308 tracked paths**, with 58 selected files inspected and 24 checksum matches. No project code or tests were executed. Historical fresh-holdout, independent-validation and promotion flags remain false; numerical replay is still open.

Prepared packets and actual submissions

The recovered September workbook contains **68 curated opportunity/partnership records** and **221 business outreach threads**. These are different views of activity, not award or customer counts. The workbook is supplemented by newly recovered primary receipts.

Work	Actual stage and preserved value
DOT NGDISM and PIDP	Three Grants.gov application chains across two opportunities: submission, validation, agency retrieval and agency tracking. An earlier schema-rejected PIDP attempt is separate.
DARPA DICE	Abstract received June 29 and reviewed July 24. Full proposal discouraged; substantial technical, budget and evidence preparation remains reusable.
NASA	Complete AI Technology Support Services quotation sent July 24; separate five-page infrastructure RFI sent July 13. No accepted quotation or award established.
DOE / ARPA-E MAGNITO	Concept paper submitted; October 2025 content/template noncompliance determination retained. Original concept/slide payload remains a recovery target.
DHS and HHS / ONC	DHS three-page Phase 1 brief/video and corrected seven-page RFI sent; HHS/ONC ten-page rendered capability packet sent. Receipt, selection and delivery remain separate.
Air Force / Navy / DLA / NSF	Air Force RFI receipt confirmed. Substantial Navy, DLA and NSF packets preserved. DLA explicitly reported MissionWeave remained In Progress, not formally submitted. Other final receipts remain unresolved.
Tradewinds / DARPA Low Resource Computing	Submitted and reviewed work retained with revision or adverse outcomes. These are pursuit assets; they are not awards or operational validation.

A June 20 readiness snapshot preserves five packages with zero local blockers and 23 portal/user gates. Later DICE receipts supersede its prepared-only state. Historical readiness is not the complete submission estate.

Federal registration is supported by a July 2026 activation notice with a July 2027 renewal date; a current portal check is separate. An executed research-collaboration agreement is retained privately.

Stage discipline: prepared, sent, received, reviewed, contracted, awarded and paid are separate. Grants.gov validation means transmission/schema processing, not a scientific, eligibility or merits determination. No government award, paid purchase order, endorsement or independently accepted result is inferred from the records summarized here.

Geometry: a larger catalog and one new screen

26 families across 11 task lanes, expanded from 18 families across nine lanes. Eight additions are brachistochrone/cycloidal descent, sunflower/phyllotaxis, Voronoi partitions, Platonic solids, gyroid nodal approximation, projected Hopf fibers, trefoil tube and superformula shell.

Existing families include mycelial and slime-mold routing, honeycomb packing, leaf veins, vascular branching, river deltas and toroidal fields. The five Platonic solids are tetrahedron, cube, octahedron, dodecahedron and icosahedron; they form one registry family. Adding candidates does not promote readiness. The June 19 frozen baseline recorded zero performance-ready families and no champion.

Fresh ideal-DC transport computation

Eight networks with 64 nodes each were computed October 4. Each uses normalized diameter 1, material volume 1 and resistivity 1; cross-section follows total wire length. All 2,016 unordered terminal pairs are equally weighted. Six graphs share terminal coordinates.

Same-terminal network	Mean conductance	Edge deletions retaining full connectivity
Square grid	0.7643	100%
Loop-augmented tree	0.7091	100%
Triangulated square grid	0.6602	100%
Minimum spanning tree	0.2553	0%
Randomized tree control	0.1971	0%
Serpentine chain	0.1849	0%

Conductance units are normalized $V/(\rho D^2)$. The square grid leads this one controlled configuration. Loops retain connectivity after a broken edge; zero full connectivity does not mean no pairs survive.

Honeycomb and sunflower conductances are 1.2564 and 0.5235, but their terminal placements differ. They are exploratory embeddings outside the controlled ranking.

Promotion remains false. This model has no time, frequency, attenuation, bandwidth, capacitance, inductance, electromagnetic waves or fluid dynamics. Contacts, manufacturing and post-failure conductance are excluded. One layout per family is not multi-scenario validation or proof of a stronger, longer-lasting real signal.

Historical research, including adverse findings

These are dated retained results, not fresh reruns in this brief.

Evidence	Recorded scope/result	Interpretation
Master Universe V2	2,172 frozen public/market series; nine models/five families; 304 harmonic winners (14.0%)	One 80/20 holdout; no universal edge established.
Hybrid Edge V7.1	80 untouched series, five outer folds; 11 screening-positive series; median MAE improvement 0%; zero robust claim gates	Neutral and adverse results retained. Surrogate and multiple-comparison controls absent.
Full Beast	Retained summary records 541,728 candidate configurations; selected baseline-relative result is adverse	Raw leaderboard closure and a historical score anomaly remain unresolved. Configurations are not independent experiments.
Phase-lock audit	1,764 candidate rows, 532 phase-lock-tagged; selected candidate ranks 557 by score and 631 by test Sharpe	Selected result underperformed its reported baseline.

Image recovery extends the original ten-photo sample: all 1,204 nonzero files from the initial 1,225-image scope were downloaded; 1,202 yielded OCR. Twenty-one zero-byte records and two unreadable image records remain explicit. Fifteen original-resolution reviews and 47 additional previews are separate from OCR. Another 241 shared-tree images remain metadata-only and unattributed. The broader Drive census contains 3,111 discovered IDs across 104 folders; neither metadata nor OCR establishes ownership, a verified experiment or complete archive coverage.

From an asset to a defensible result

1. Identify source, version, owner, dependencies and the exact claim.
2. Freeze authorized inputs, comparator, budget, constraints and primary metric.
3. Use appropriate holdouts, uncertainty and search controls.
4. Preserve outputs, failures, environment and reproducible commands.
5. Obtain independent or representative validation before operational claims.
6. Connect accepted results to buyer-approved economics and actual terms.

Release gate: publishing a source or demonstration does not certify the research, establish customer savings or authorize production control. This review expands the documented estate without converting breadth into unsupported performance or price.

[lumen-core.ai](#) · [Public repository](#) · [Reviewer start](#)

Sources: canonical implementation and readiness records; geometry registry; June 18 research audit; October 4 DC model; expanded trading/engineering inventories; private receipt and attachment index with sanitized public stages. Original emails, identifiers and agreement details remain private.